

ExoNet: Project Proposal

ITAI 2372 Final Project, Implementation Track

Trilok, Group 01 (solo). April 2026.

1. Problem

NASA's Kepler mission watched more than 150,000 stars looking for planets passing in front of them. It produced thousands of transit-like signals. Each signal needs to be vetted to figure out whether it is really a planet, or a false positive caused by something else (an eclipsing binary star, instrument noise, etc.). Vetting by hand takes a lot of expert time, and missions like TESS keep adding more candidates to the queue.

I want to see if a small machine learning model can learn that vetting decision from the public KOI table well enough to be useful as a first-pass filter. The model is not meant to replace astronomers, just to triage the easy cases.

2. Proposed Solution

A supervised classifier (Random Forest) that takes ten KOI features covering the orbit, the candidate planet itself, and the host star, and predicts CONFIRMED vs FALSE POSITIVE. The deliverable is a small Python CLI with a train command and a predict command, plus tests.

I picked Random Forest over a neural network because the dataset is small (about 7,500 usable rows) and tabular. Trees handle non-linear interactions, do not need feature scaling, and give feature importances I can talk about in the slides. A neural net would be harder to defend in a presentation and would not obviously do better at this scale.

3. Data

The dataset is the Kepler Object of Interest (KOI) cumulative table from the NASA Exoplanet Archive. After filtering out CANDIDATE rows (label undecided) there are about 7,500 rows split between CONFIRMED and FALSE POSITIVE. The code caches the CSV locally on first run.

If the archive is not reachable when grading, the data loader prints a warning and falls back to a synthetic dataset so the rest of the code still runs. Metrics from a synthetic run are obviously not scientifically meaningful, but the pipeline is still demonstrable.

4. Methodology

- Load the KOI cumulative table and cache it to disk.
- Coerce features to numeric, drop empty rows, fill remaining NaNs with the column median.
- Drop CANDIDATE rows. Map CONFIRMED to 1 and FALSE POSITIVE to 0.

- Stratified 80/20 train/test split with a fixed seed for reproducibility.
- Fit RandomForestClassifier (300 trees, balanced class weights).
- Score on the test set with accuracy, precision, recall, F1, and ROC-AUC.
- Save the model with joblib so the predict command can reload it.

5. Success Criteria

- End-to-end pipeline runs from a single command on a clean install.
- Test-set accuracy of at least 0.85 and ROC-AUC of at least 0.92 on the real KOI data.
- All unit tests pass.
- README is clear enough that a classmate can run the project in five minutes or less.

6. Timeline

Five weekly checkpoints. The week numbers are relative to when I started, not calendar weeks.

Week	Focus	Output
1	Pick topic and dataset	This proposal
2	Explore the data, decide which columns to use	Notes, data_dictionary.md
3	First working model and metrics harness	train.py, evaluate.py, baseline numbers
4	Tune hyperparameters and write tests	tests/, README usage examples
5	Documentation, slides, polish	Final repo and slides

7. Risks and What I Plan to Do About Them

A few things could go wrong, with how I am planning to handle each:

- NASA archive download fails. There is a synthetic-data fallback so the code at least runs.
- Class imbalance. CONFIRMED and FALSE POSITIVE are not perfectly balanced, so I use a stratified split and `class_weight="balanced"` in the Random Forest.
- Model overfits. 300 trees with `min_samples_leaf=2` helps, and the hold-out test set will catch obvious overfitting.
- Running out of time. Plan is to lock the feature list at end of week 3 and not touch it again, even if I think I have a better idea.

8. Deliverables

- GitHub repo named `FP_GROUP01_Trilok_ITAI2372` with code, tests, and docs.
- `README.md` with run instructions.
- `requirements.txt` for reproducible installs.
- This proposal (`docs/proposal.pdf`).
- Testing plan (`docs/testing_plan.pdf`).
- Final presentation slides (`presentation.pdf`).

9. Citation

This research has made use of the NASA Exoplanet Archive, which is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program.